



The potential of traditional herbal active ingredients in the treatment of sarcopenia animal models: focus on therapeutic effects and mechanisms

Kaixi Ding¹ · Wei Jiang^{1,2} · Juejue Zhangwang^{1,2} · Yu Wang³ · Jing Zhang² · Ming Lei²

Received: 3 July 2023 / Accepted: 18 July 2023 / Published online: 1 August 2023

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2023

Abstract

Sarcopenia is a major global public health problem that harms individual physical function. In 2018, the European Working Group on Sarcopenia in the Elderly 2 classified sarcopenia into primary and secondary sarcopenia. However, information on the pathogenesis and effective treatment of primary and secondary sarcopenia is limited. Traditional herbal active ingredients have biological activities that promote skeletal muscle health, showing potential preventive and therapeutic effects on sarcopenia. Therefore, this narrative review aims to provide a comprehensive overview of global traditional herbal active ingredients' beneficial therapeutic effects and molecular mechanisms on sarcopenia-related animal models. For this purpose, we conducted a literature search in three databases, PubMed, Web of Science, and Embase, consistent with the review objectives. After the screening, 12 animal studies met the review themes. The review results showed that the pathological mechanisms in sarcopenia-related animal models include imbalanced protein metabolism, oxidative stress, inflammation, apoptosis, insulin resistance, endoplasmic reticulum stress, impaired mitochondrial biogenesis, and autophagy-lysosome system aggravation. Eleven traditional herbal active ingredients exerted positive anti-sarcopenic effects by ameliorating these pathological mechanisms. This narrative review will provide meaningful insight into future studies regarding traditional herbal active ingredients for treating sarcopenia.

Keywords Sarcopenia · Herbal medicine · Phytochemicals · Therapeutic effect · Molecular mechanisms of pharmacological action

Abbreviations

AI	Attractylenolide I	BMP	Bone morphogenetic protein
Akt	Protein kinase B	CHOP	C/EBP homologous protein
ALK	Activin receptor-like kinase	CK	Ginsenoside compound K
ATF6	Activating transcription factor 6	CKD	Chronic kidney disease
Bax	Bcl2-associated X protein	COPD	Chronic obstructive pulmonary disease
Bcl2	B-cell lymphoma 2 protein	CS	Carnosol
		DCS	Dimethyl-carnosol
		DCSD	Dimethyl-carnosol-D6
		DEXA	Dexamethasone
		DHM	Dihydromyricetin
		DMF	5,7-Dimethoxyflavone
		ER	Endoplasmic reticulum
		FOXO3a	Forkhead box class O 3a
		GPx	Glutathione peroxidase
		GR	Glutathione reductase
		GSK-3β	Glycogen synthase kinase 3β
		IGF-1	Insulin-like growth factor 1
		IKB	Inhibitor of κB
		IKK	IκB kinase
		JAK	Janus kinase

Kaixi Ding and Wei Jiang contributed equally to this work.

✉ Jing Zhang
zhangjing@cdutcm.edu.cn

✉ Ming Lei
minglei2828@126.com

¹ School of Clinical Medicine, Chengdu University of Traditional Chinese Medicine, Chengdu 610075, China

² Hospital of Chengdu University of Traditional Chinese Medicine, Chengdu 610075, China

³ Jiangsu Provincial Hospital of Traditional Chinese Medicine, Nanjing 210029, China

LPS	Lipopolysaccharides
MDA	Malondialdehyde
MHC	Major histocompatibility complex
mTOR	Mammalian target of rapamycin
MSTN	Myostatin
MuRF1	Muscle RING-finger protein-1
MyHC	Myosin heavy chain
MyoD	Myogenic differentiation antigen
NF- κ B	Nuclear factor-kappa B
NRF-1	Nuclear factor erythroid 2 related factor 1
ORX	Orchiectomized
Pax7	Paired box protein Pax-7
PGC-1 α	Proliferator-activated receptor gamma coactivator 1-alpha
PI3K	Phosphoinositide 3-kinases
PKM2	Pyruvate kinase isozyme M2
S-Rg3	20(S)-Ginsenoside Rg3
SNAP23	Synaptosome-associated protein 23
STAT	Signal transducer and activator of transcription
TFAM	Mitochondrial transcription factor A
TNF- α	Tumor necrosis factor-alpha
WOS	Web of Science
XIAP	X-linked inhibitor of apoptosis protein

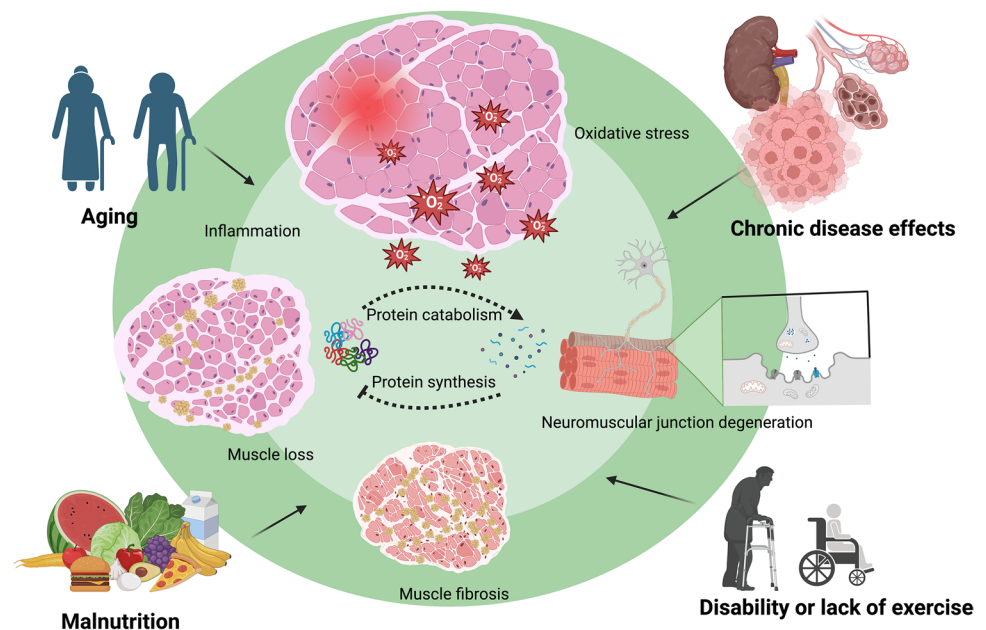
Introduction

As societies age globally, sarcopenia is becoming a significant disease burden and health problem worldwide (Bruyère et al. 2019; Traub et al. 2020). Sarcopenia is a syndrome typically characterized by a progressive decrease in muscle strength, muscle mass, and muscle quantity (Cruz-Jentoft et al. 2019). Risk factors such as aging, systemic chronic diseases, physical inactivity, and poor diet can trigger and exacerbate sarcopenia (Cruz-Jentoft et al. 2019, 2017; Mijnders et al. 2016). Sarcopenia often occurs in the geriatric period. However, it can also occur earlier in human life (Cruz-Jentoft et al. 2019). According to a 2022 meta-analysis study, the global prevalence of sarcopenia ranged from 8 to 36% in individuals under 60 and from 10 to 27% in individuals over 60 (Petermann-Rocha et al. 2022). The results of another 2019 meta-analysis study showed that the prevalence of sarcopenia in older age groups was higher in nursing homes (51% for men and 31% for women) and hospitals (23% for men and 24% for women) (Papadopoulou et al. 2021). With the progressive loss of muscle function, patients with sarcopenia are at risk for many adverse events, including falls, fractures, progressive reduction in activities of daily living, loss of independence, cardiopulmonary disease, cognitive impairment, and death (Cruz-Jentoft et al. 2019). Healthcare costs are higher for sarcopenic patients than for non-sarcopenic patients (Bruyère et al. 2019). Direct healthcare costs for older adults with sarcopenia are twice as

significant as those without (Steffl et al. 2017). In addition, sarcopenia raises the risk of hospitalization and the associated costs of hospitalization and care during hospitalization (Cawthon et al. 2017). Predictably, the health impact and economic burden of sarcopenia will continuously increase. Therefore, focusing on the pathogenesis of sarcopenia and developing practical, economical, and safe treatments for it are of great social importance and benefit to human health.

Sarcopenia divides into primary and secondary, with common and different pathogenesis (Bauer et al. 2019; Dhillon and Hasni 2017). Generally, the histopathological features of both primary and secondary sarcopenia are similar, with early manifestations of muscle size reduction and progressive manifestations of muscle fiber replacement by fat, increased fibrosis, oxidative stress, altered muscle metabolism, and neuromuscular junction degeneration (Fig. 1) (Dhillon and Hasni 2017). Primary sarcopenia is defined explicitly as resulting from aging. Aging-related exercise deficiency, hormonal and cytokine imbalance, accumulating non-contractile dysfunctional proteins in skeletal muscle, failure of satellite cell activation, and genetic aging are essential to primary sarcopenia (Dhillon and Hasni 2017). The main causative factors of secondary sarcopenia include systemic diseases, lack of exercise, and malnutrition (Cruz-Jentoft et al. 2019). The mechanisms of secondary sarcopenia vary from disease to disease. For example, in chronic obstructive pulmonary disease (COPD), changes in several cytokines (IL-6, muscle growth inhibitor, fibronectin type III domain-containing protein 5) and signaling pathways (RANKL/RANK) are involved in muscle-skeleton crosstalk, leading to muscle dysfunction (Ma et al. 2022); in chronic kidney disease (CKD), adverse factors such as chronic low-grade inflammation, metabolic acidosis, insulin resistance, vitamin D deficiency, and hormonal disorders promote protein degradation and reduce protein synthesis, resulting in a negative protein balance, which in turn causes muscle loss (Sabatino et al. 2021). Lack of exercise or muscle wasting causes muscle atrophy by reducing the secretion of myocytokines, disrupting the balance between anti-inflammatory and pro-inflammatory mediators, and affecting protein metabolism (Barbalho et al. 2020). An inactive lifestyle also accelerates the development of aging-associated sarcopenia (Naseeb and Volpe 2017). Low inflammation, oxidative stress, energy deficiency, disturbances in protein anabolism, and impaired cell membrane repair are essential mechanisms in malnutrition-associated sarcopenia (Papadopoulou et al. 2021; Sieber 2019). These mechanisms also contribute to aging-associated sarcopenia (Papadopoulou et al. 2021; Sieber 2019).

Many natural active ingredients have anti-inflammatory, anti-fatigue, muscle damage prevention, muscle atrophy prevention, and muscle regeneration and differentiation bioactivities, promoting skeletal muscle health (Rondanelli

Fig. 1 Risk factors and pathogenesis of sarcopenia

et al. 2016). Flavonoid compounds such as isobavachalcone, delphinidin, daidzein, and quercetin are involved in the regulation of protein turnover, mitochondrial activity, and myogenesis and exert anti-muscular atrophy effects (Kim and Hwang 2020a). Some plant-derived natural products, including curcumin, resveratrol, catechins, and soy protein, have beneficial effects on sarcopenia by increasing muscle mass and strength, inhibiting muscle atrophy, regulating anabolism, and enhancing mitochondrial function (Bagherniya et al. 2022). In addition, traditional herbal active ingredients have been considered potential therapeutic options for treating sarcopenia.

Traditional herbal active ingredients have therapeutic effects on many human diseases, such as hypertension, malaria, cancer, cardiovascular disease, diabetes, and sarcopenia (Bagherniya et al. 2022; Takemoto et al. 2017; Yuan et al. 2016). Many traditional herbal active ingredients such as atractylenolide I, curcumin, and carnosol exerted excellent therapeutic effects and mechanisms in various animal models related to sarcopenia. Two previous reviews focused on the therapeutic effects and molecular mechanisms of flavonoids and plant-derived natural products in animal models of aging-related sarcopenia (Bagherniya et al. 2022; Kim and Hwang 2020a). No review has examined traditional herbal active ingredients' therapeutic effects and molecular mechanisms in secondary and primary sarcopenia animal models. Given the many similarities in the pathogenesis of secondary and primary sarcopenia, therapeutic mechanism studies of secondary sarcopenia may provide helpful insights for those of primary sarcopenia (Dhillon and Hasni 2017; Wiedmer et al. 2021). Compared to the two previous reviews, the present

review differs in two ways: (1) It focuses on more traditional herbal medicinal active ingredients and (2) it also addresses the therapeutic effects and mechanisms of traditional herbal medicinal active ingredients of some secondary sarcopenia-related animal models, in addition to primary sarcopenia-related animal models. In the following, we comprehensively summarize 11 traditional herbal active ingredients (baicalin, atractylenolide I, carnosol, ginsenoside Rg3, ginsenoside compound K, curcumin, icariin, 5,7-dimethoxyflavone, morroniside, dihydromyricetin, glabridin) on the therapeutic effects and molecular mechanisms in primary and secondary sarcopenia-related animal models (Fig. 2). The present review will provide a comprehensive and in-depth molecular and therapeutic perspective for subsequent studies on traditional herbal active ingredients for treating sarcopenia.

Research methodology

This narrative review aimed to explore the potential role and therapeutic mechanisms of world traditional herbal active ingredients in sarcopenia-related animal models. A literature search was conducted in three databases: PubMed, Embase, and Web of Science. The search terms included the following: “sarcopenia,” “animal models,” “therapeutic effects,” “therapeutic mechanisms,” “traditional herbal medicines,” and “active ingredients.” Relevant literature was retrieved with filters not turned on. In addition, the literature search was expanded by hand search retrieval. Candidates for the target study should meet the following criteria:

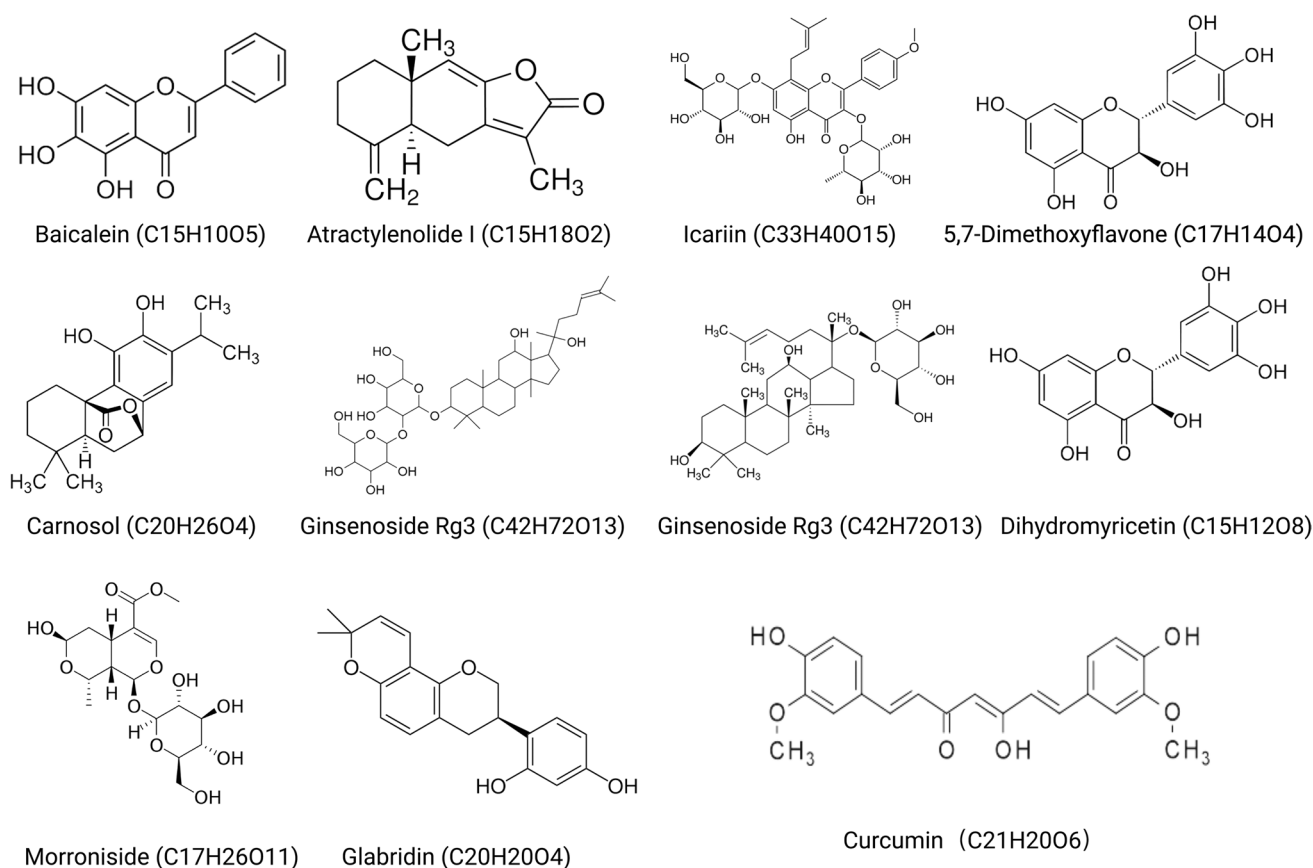


Fig. 2 The names and chemical formulas of 11 traditional herbal active ingredients

- (1) Active ingredients of world traditional herbs as interventions
- (2) Use of animal models related to primary or secondary sarcopenia
- (3) Studies including therapeutic effects and molecular mechanisms

Twelve original studies were ultimately included in this narrative review (Table 1). Cumulatively, 11 different traditional herbal active ingredients were used in the 12 original studies.

Therapeutic effects and molecular mechanisms of traditional plant medicine active ingredients on sarcopenic animal models

Baicalin, atractylenolide I, carnosol, and its analogs

Baicalin (5,6-dihydroxy-7-O-glucuronide flavone) (C₂₁H₁₈O₁₁) is a lipophilic flavonoid glycoside isolated from the traditional plant *Scutellaria baicalensis* Georgi. It has

anti-inflammatory, antioxidant, and anti-apoptotic biological activities. Baicalin can regulate multiple signaling pathways, including IκB kinase (IKK)/inhibitor of κB (IκB)/NF-κB, phosphoinositide 3-kinases (PI3K)/AKT/FOXO1, and Janus kinase (JAK)/signal transducer and activator of transcription (STAT), and exert therapeutic effects on ulcerative colitis, cancer, depressive behavior, and myocardial ischemia-reperfusion injury (Guo et al. 2019; Shen et al. 2019; Singh et al. 2021; Xu et al. 2020). Atractylenolide I, a lactone compound (C₁₅H₁₈O₂), is the main active ingredient of the traditional drug *Rhizoma Atractylodes macrocephalae*. It has anti-cancer, anti-inflammatory, and neuroprotective activities and has therapeutic potential for rectal colon cancer, melanoma, depression, gastric cancer, chronic atrophic gastritis, and atopic dermatitis (Deng et al. 2021). Carnosol (C₂₀H₂₆O₄) is a diterpenoid extracted from *Rosmarinus officinalis*, a natural spice plant with antioxidant, anti-inflammatory, and anti-cancer pharmacological effects (Lu et al. 2021). Carnosol regulates various signaling pathways, including Akt/mammalian target of rapamycin (mTOR), Bcl2-Associated X protein (Bax)/B-cell lymphoma 2 protein (Bcl2), and NF-κB, STAT3-6, and is broadly used in treating cardiovascular diseases, neurodegenerative diseases, and cancer

Table 1 General information of the 12 included studies

Title	Year	Authors	Traditional Herbs	Regional and national locations of traditional herbs	Traditional herbal active ingredients	Traditional medicine category	Abbreviation of journal title	Doi	References
Baicalin, a component of <i>Scutellaria baicalensis</i> , alleviates anorexia and inhibits skeletal muscle atrophy in experimental cancer cachexia	2014	Bin Li et al	<i>Scutellaria baicalensis</i> Georgi	China, Japan, and Korea	Baicalin	Traditional Chinese medicine	<i>Tumour Biol</i>	10.1007/s13277-014-2558-9	(B. Li et al. 2014)
Carnosol and its analogues attenuate muscle atrophy and fat lipolysis induced by cancer cachexia	2021	Shanshan Lu et al	<i>Rosmarinus officinalis</i>	India and Mediterranean region	Carnosol and its analogs	Indian traditional medicine and folk medicine	<i>J Cachexia Sarcopenia Muscle</i>	10.1002/jcsm.12710	(Lu et al. 2021)
Atractylenolide I ameliorates cancer cachexia through inhibiting biogenesis of IL-6 and tumour-derived extracellular vesicles	2022	Meng Fan et al	<i>Rhizoma Atractylodes Macrocephalae</i>	China, Japan, and Korea	Atractylenolide I	Traditional Chinese medicine	<i>J Cachexia Sarcopenia Muscle</i>	10.1002/jcsm.13079	(Fan et al. 2022a)
20(S)-Ginsenoside Rg3 promotes myoblast differentiation and protects against myotube atrophy via regulation of the Akt/mTOR/FOXO3 pathway	2020	Manying Wang et al	<i>Panax ginseng</i> CA Meyer, <i>Panax notoginseng</i> , and <i>Panax quinquefolius</i> L.	China, Korea, North America, and Russia	20(S)-Ginsenoside Rg3	Chinese, Japanese, and Korean traditional medicine	<i>Biochem Pharmacol</i>	10.1016/j.bcp.2020.114145	(M. Wang et al. 2020a, b)

Table 1 (continued)

Title	Year	Authors	Traditional Herbs	Regional and national locations of traditional herbs	Traditional herbal active ingredients	Traditional medicine category	Abbreviation of journal title	Doi	References
Ginsenoside compound K ameliorates palmitate-induced atrophy in C2C12 myotubes via pro-myogenic effects and AMPK/autophagy-mediated suppression of endoplasmic reticulum stress	2022	Tae Jin Kim et al	<i>Panax ginseng</i> CA Meyer, <i>Panax notoginseng</i> , and <i>Panax quinquefolius</i> L	China, Korea, North America, and Russia	Ginsenoside compound K	Chinese, Japanese, and Korean traditional medicine	<i>J Ginseng Res</i>	10.1016/j.jgr.2021.09.002	(Kim et al. 2022)
Curcumin ameliorates CKD-induced mitochondrial dysfunction and oxidative stress through inhibiting GSK-3 beta activity	2020	Dongtao Wang et al	<i>Curcuma longa</i> species	China, and India	Curcumin	Traditional Chinese medicine	<i>J Nutr Biochem</i>	10.1016/j.jnutbio.2020.108404	(D. Wang et al. 2020a, b)
Curcumin treatment prevents increased proteasome and apoptosis activities in rat skeletal muscle during reloading and improves subsequent recovery	2012	Emilie Vazeille et al	<i>Curcuma longa</i> species	China, and India	Curcumin	Traditional Chinese medicine	<i>J Nutr Biochem</i>	10.1016/j.jnutbio.2010.11.021	(Vazeille et al. 2012)
A phyto molecule icariin protects from sarcopenia partially by suppressing myosin heavy chain degradation in orchietomized rats	2022	Wen-Yao Yang et al	<i>Epimedium</i>	China, Japan, and Korea	Icariin	Traditional Chinese medicine	<i>Adv Biol (Weinh)</i>	10.1002/adbi.202200162	(Yang et al. 2022)

Table 1 (continued)

Title	Year	Authors	Traditional Herbs	Regional and national locations of traditional herbs	Traditional herbal active ingredients	Traditional medicine category	Abbreviation of journal title	Doi	References
The 5,7-dimethoxyflavone suppresses sarcopenia by regulating protein turnover and mitochondria biogenesis-related pathways	2020	Changhee Kim et al	<i>Kaempferia parviflora</i>	Thailand, Cambodia, and Vietnam	5,7-Dimethoxyflavone	Traditional Thai medicine	<i>Nutrients</i>	10.3390/nu12041079	(Kim and Hwang 2020b)
Morroniside ameliorates inflammatory skeletal muscle atrophy via inhibiting canonical and non-canonical NF- κ B and regulating protein synthesis/degradation	2022	Xiangjiao Yi et al	<i>Cornus officinalis</i>	China, Japan, and Korea	Morroniside	Traditional Chinese medicine	<i>Front Pharmacol</i>	10.3389/fphar.2022.1056460	(Yi et al. 2022)
Dihydromyricetin attenuates dexamethasone-induced muscle atrophy by improving mitochondrial function via the PGC-1 α pathway	2018	Yujie Huang et al	<i>Ampelopsis grossedentata</i>	China	Dihydromyricetin	Chinese, Indian, and Japanese traditional medicine	<i>Cell Physiol Biochem</i>	10.1159/000493040	(Y. Huang et al. 2018a, b)
Glabridin inhibits dexamethasone-induced muscle atrophy	2019	Yasukiyo Yoshioka et al	<i>Glycyrrhiza glabra</i> L.	China, Iran, Mediterranean region, and India	Glabridin	European, Asian traditional medicine	<i>Arch Biochem Biophys</i>	10.1016/j.abb.2019.02.006	(Yoshioka et al. 2019)

research (Kashyap et al. 2017). More recently, researchers have explored the potential of these three components for treating cachexia-related sarcopenia muscle.

In a study, Li et al. found that baicalin inhibited the reduction in weight and diameter of the gastrocnemius muscle in CT26 adenocarcinoma-induced cachexia mice (Li et al. 2014). Furthermore, the expression of muscle RING-finger protein-1 (MuRF1) and atrogin-1, specific proteins associated with skeletal muscle atrophy, was inhibited (Li et al. 2014). Nuclear factor-kappa B (NF- κ B) activation promotes muscle atrophy in tumor-bearing mice and cancer patients (Acharyya et al. 2005; Rhoads et al. 2010). Intervention with baicalin decreased phosphorylated p65 levels and increased I κ B α levels, suggesting it inhibited NF- κ B activation in CT26 adenocarcinoma-induced cachexia mice (Li et al. 2014). In another study, Lu et al. screened and found that carnosol (CS) and its two analogs, dimethyl-carnosol (DCS) and dimethyl-carnosol-D6 (DCSD), were able to improve myotubular atrophy by inhibiting inflammatory cytokine-induced activation of NF- κ B in C2C12 myotubes, activating Akt signaling, and increasing the expression of myogenic differentiation antigen (MyoD) and major histocompatibility complex (MHC) (Lu et al. 2021). CS and DCSD improved the cross-sectional area of gastrocnemius muscle fibers in C26 tumor-bearing mice (Lu et al. 2021). CS and DCSD treatments modulated NF- κ B and Akt signaling pathways and inhibited atrogin-1 expression, affecting protein synthesis and degradation in the muscle of cancer C26 tumor-bearing mice (Lu et al. 2021). Fan et al. investigated the role and mechanism of atractylenolide I (AI) on cachexia-associated muscle atrophy by regulating cachexia factors, IL-6, and extracellular vesicles released by cancer cells (Fan et al. 2022a). In vivo, AI significantly inhibited the reduction of myofiber cross-sectional area in the anterior tibialis muscle of C26 tumor-bearing mice and down-regulated the levels of IL-6 and extracellular vesicles (Fan et al. 2022a). In vitro, Fan et al. found that STAT3 gene overexpression counteracted the up-regulation of MHC and MyoD expression and the downregulation of atrogin-1 and MuRF1 expression by AI in C26 cells (Fan et al. 2022a). In contrast, STAT3 knockdown slightly enhanced the up-regulation of MyoD expression and the downregulation of atrogin-1 and MuRF1 expression levels by AI (Fan et al. 2022a). These results suggest that activation of STAT3 may promote cachexia-associated muscle atrophy. According to a previous study by Fan et al., the STAT3/pyruvate kinase isozyme M2 (PKM2)/synaptosome-associated protein 23 (SNAP23) pathway affected aerobic glycolysis, promoted extracellular vesicle biogenesis, and induced cancer cachexia (Fan et al. 2022b). By inhibiting STAT3/PKM2/SNAP23 signaling, AI reduced IL-6 levels and extracellular vesicle generation, exerting an anti-cachexia-associated muscle atrophy effect (Fan et al. 2022a).

Ginsenoside Rg3 and ginsenoside compound K

Ginsenoside Rg3 (C₄₂H₇₂O₁₃) is a saponin isolated from several well-known Asian herbs, *Panax ginseng* CA Meyer, *Panax notoginseng*, and *Panax quinquefolius* L (Wang et al. 2022). It has therapeutic effects on diseases such as tumors, obesity, diabetes, hypertension, lung inflammation, neuroinflammation, and brain and myocardial ischemia-reperfusion injury (Lee et al. 2020; Ma et al. 2022; Wang et al. 2022). Ginsenoside compound K (CK), also known as proto-panaxadiol 20-O-glucoside (C₃₆H₆₂O₈), is a deglycosylated metabolite of various ginsenosides. CK is a rare ginsenoside transformed by intestinal bacteria and has stronger biological activity than other ginsenosides (Liu et al. 2022). CK has a variety of beneficial biological activities, including anti-cancer, anti-inflammatory, anti-diabetic, anti-angiogenic, anti-aging, neuroprotective, and hepatoprotective (Liu et al. 2022). Previous studies have found that ginseng extracts (rich in ginsenoside Rb1, ginsenoside Rb2) and ginsenoside Rg1 have improved muscle atrophy in vitro and in vivo (Go et al. 2017; Li et al. 2017; Seok et al. 2021). Ginsenoside Rg3 and CK also have anti-muscular dystrophy therapeutic effects.

In vitro, Wang et al. found that 20(S)-ginsenoside Rg3 (S-Rg3) more effectively promoted Myf5 expression in dexamethasone (DEXA)-induced atrophic C2C12 myotubes compared to ginsenoside Rb3 and Rg1 (Wang et al. 2020a, b). Myogenic cells expressing Myf5 facilitate replenishing muscle stem cells, which repair skeletal muscle (Günther et al. 2013). In addition, S-Rg3 dose-dependently promoted MHC and myostatin (MSTN) expression in C2C12 myogenic cells and promoted myogenic differentiation (Wang et al. 2020a, b). Mechanistically, S-Rg3 promoted DEXA-induced activation of the Akt/mTOR pathway in C2C12 myotubes and reversed the downstream FOXO3 nuclear translocation (Wang et al. 2020a, b). In addition, S-Rg3 dose-dependently promoted MHC and MSTN expression in C2C12 myogenic cells and promoted myogenic differentiation (Wang et al. 2020a, b). Mechanistically, S-Rg3 promoted DEXA-induced activation of the Akt/mTOR pathway in C2C12 myotubes and reversed the downstream FOXO3 nuclear translocation (Wang et al. 2020a, b). Furthermore, Wang et al. used an Akt inhibitor, GDC-0068, to pre-incubate C2C12 myogenic cells, after which the expression of myostatin and Myf5 and the phosphorylation levels of Akt and mTOR were reduced (Wang et al. 2020a, b). In contrast, the nucleoplasmic translocation of FOXO3 downstream of the Akt/mTOR pathway was not significantly increased in C2C12 myogenic cells treated with the combination of S-Rg3 and GDC-0068, suggesting that the promotion of myogenic differentiation by S-Rg3 is dependent on the activation of the Akt/mTOR/FOXO3 pathway (Wang et al. 2020a, b).

In addition, DEXA induced myotubular atrophy in C2C12 and increased the expression of two muscle atrophy proteins, MuRF1, and atrogin-1 (Wang et al. 2020a, b). S-Rg3 reversed these changes and exerted anti-DEXA-induced muscle atrophy effects. Evidence supports the involvement of the Akt/mTOR/FOXO3 pathway in the inhibitory effect of ginsenoside Rg1 on atrophic muscles (Li et al. 2017). S-Rg3 did not increase myotube diameter in GDC-0068-pretreated DEXA-induced myotube atrophy, suggesting that the protective effect of S-Rg3 on myotube atrophy may also rely on the activation of the Akt/mTOR/FOXO3 pathway [104]. In vivo, S-Rg3 significantly maintained the percentage of muscle area and increased climbing ability in *Drosophila*. S-Rg3 promoted dFOXO nucleoplasmic translocation, counteracting muscle atrophy in *Drosophila* (Wang et al. 2020a, b). In another animal study, Kim et al. investigated CK's therapeutic effects and mechanisms in a mice model of obesity-related muscular atrophy (Kim et al. 2022). CK reversed the reduction in myotube thickness and up-regulation of cleaved caspase-3 expressions and increased caspase-3 activities in mice with high-fat diet-induced obesity sarcopenia (Kim et al. 2022). In addition, CK supplementation increased MyH4 (a marker of type IIb “fast twitch” fibers) expression; however, it did not recover the decrease in MyH7 (a marker of type I “slow muscle” fibers) expression (Kim et al. 2022). It has been shown that endoplasmic reticulum (ER) stress and apoptosis exist in skeletal muscle and palmitate-induced C2C12 myocytes in type 2 diabetic rats (Bryner et al. 2012; Reddy et al. 2018). Moreover, amelioration of ER stress attenuates palmitate-induced apoptosis in myotubular cells (Woodworth-Hobbs et al. 2017). Following such known information, in vivo, Kim et al. experimentally found that CK attenuated ER stress in the skeletal muscle of sarcopenic obese mice, as evidenced by reduced eIF2 α phosphorylation, activating transcription factor 6 (ATF6) cleavage, and C/EBP homologous protein (CHOP) expression (Kim et al. 2022). AMPK can reduce ER stress and apoptosis by promoting autophagy-dependent signaling (Pyun et al. 2021). In vitro and in vivo, CK dose-dependently increased AMPK phosphorylation, autophagy levels, and autophagosome numbers (Kim et al. 2022). In vitro, CK's ER stress-mitigating, apoptosis-reducing, and myotube formation-promoting effects were inhibited when palmitate-induced C2C12 myotubes were pretreated with 3-MA, an autophagy inhibitor (Kim et al. 2022). These results show that the AMPK/autophagy signaling pathway may be involved in the effects of CK on ER stress, apoptosis, and anti-muscle atrophy. In addition, in vivo, CK increased the phosphorylation of p38 and Akt in mice skeletal muscle and increased the levels of several myogenic markers, MyoD, MHC, and myogenin, demonstrating the potential myogenic effects of CK (Kim et al. 2022).

Curcumin

Curcumin (C₂₁H₂₀O₆) is a flavonoid from the traditional herb *Curcuma longa* species. Curcumin has anti-inflammatory, antioxidant, anti-cancer, and metabolism-improving biological properties. It has therapeutic potential for metabolic syndrome, rheumatoid arthritis, ulcerative colitis, and Alzheimer's disease (Akaberi et al. 2021). Curcumin also has the potential to benefit CKD-related and immobilization-induced muscle atrophy.

Patients with CKD usually experience a significant loss of muscle mass and motor ability later in the disease (Roshanravan et al. 2017). In CKD, uremia-related inflammation, oxidative stress, and insulin resistance promote skeletal muscle dysfunction (Roshanravan et al. 2017). Mitochondrial dysfunction caused by insulin resistance disrupts fatty acid metabolism in myocytes (Befroy et al. 2007). Furthermore, oxidative stress caused by mitochondrial muscle dysfunction accelerates protein hydrolysis in muscle atrophy (Wang et al. 2018a, b). Therapies targeting mitochondrial dysfunction and oxidative stress have shown the potential to improve CKD-related muscle atrophy and decreased motor performance (Wang et al. 2018a, b). Wang and his team found that intake of a curcumin diet reversed the decrease in mass and fiber cross-sectional area of the flounder and gastrocnemius muscles in CKD mice. In addition, curcumin increased grip strength and running distance in CKD mice (Wang et al. 2020a, b). Curcumin improved mitochondrial function, mitochondrial ETC enzyme activity, and mitochondrial biosynthesis in mice skeletal muscle, as evidenced by decreased mitochondrial O₂^{•-} production, increased mitochondrial ATP levels and mitochondrial oxygen consumption rate, increased mitochondrial electron transport chain complex enzyme activity, and increased mitochondrial transcription factor A (TFAM), proliferator-activated receptor gamma coactivator 1-alpha (PGC-1 α), and nuclear factor erythroid 2-related factor 1 (NRF-1) expression levels, respectively (Wang et al. 2020a, b). Curcumin increased superoxide dismutase, GSH, glutathione peroxidase (GPx), and glutathione reductase (GR) levels, decreased malondialdehyde (MDA) levels, and restored muscle mitochondrial redox homeostasis in CKD mice (Wang et al. 2020a, b). Knockdown of glycogen synthase kinase 3 β (GSK-3 β) promotes increased mitochondrial biogenesis and improves oxidative metabolism in skeletal muscle (Theeuwes et al. 2017). In vivo, Wang et al. demonstrated that GSK-3 β knockdown also ameliorated CKD-induced mitochondrial dysfunction and reduced mitochondrial electron transport chain complex enzyme activity and oxidative stress in skeletal muscle, which was beneficial for CKD-induced muscle atrophy (Wang et al. 2020a, b). It can be seen that GSK-3 β knockdown and curcumin exert similar effects on CKD-induced muscle atrophy and mitochondrial dysfunction. In vitro and in vivo, curcumin

inhibited the elevated levels of GSK-3 β activity in CKD mice skeletal muscle and H₂O₂-induced C2C12 myotubes (Wang et al. 2020a, b). These results strongly suggest that the effect of curcumin on CKD-induced muscle atrophy is likely to arise from its mechanism of mitochondrial biogenesis promotion, mitochondrial function improvement, and oxidative stress inhibition through the inhibition of enhanced GSK-3 β activity.

The presence of inflammatory processes and activation of oxidative stress pathways leads to pathological changes such as muscle damage, apoptosis, and accelerated protein hydrolysis in fixation-induced wasting muscle atrophy (Bar-Shai et al. 2008; Vazeille et al. 2012). Considering curcumin's anti-inflammatory and antioxidant activity and its high tolerance, Vazeille et al. explored the therapeutic effects and molecular mechanisms of curcumin in rats with immobilization-induced muscle atrophy (Vazeille et al. 2012). Intraperitoneal administration of curcumin improved muscle atrophy and reduced muscle fiber cross-sectional area in the affected gastrocnemius muscle between the immobilization and recovery periods in rats (Vazeille et al. 2012). The ubiquitin-proteasome system is primarily responsible for the breakdown of contractile skeletal muscle proteins. Excessive activation of the ubiquitin-proteasome system can lead to muscle atrophy (Attaix et al. 2005). Normalization of ubiquitin-proteasome system activity is beneficial in increasing muscle mass in wasted muscle atrophy (Vazeille et al. 2012). Vazeille et al. found that curcumin treatment inhibited the enhancement of proteasome pancreatic rennet-like activity and increased protein ubiquitination rate in the skeletal muscle of immobilized mice, which may facilitate skeletal muscle regeneration (Vazeille et al. 2012). In addition, curcumin also modulated the mitochondrial apoptotic pathway, as evidenced by increased levels of anti-apoptotic X-linked inhibitor of apoptosis protein (XIAP) during recovery, decreased levels of pro-apoptotic Smac/Diablo protein during stabilization, and decreased caspase-9 activity associated with apoptotic vesicles during recovery (Vazeille et al. 2012). Overall, curcumin may regulate apoptosis in muscle during immobilization and promote the initial steps of muscle regeneration.

Icariin

Icariin (C₃₃H₄₀O₁₅), a flavonol component, is mainly extracted from the stem and leaves of the traditional plant *Epimedium* (Ding et al. 2022). Icariin has shown therapeutic effects on neurodegenerative diseases, cardiovascular diseases, osteoporosis, depression, tumors, and diabetes (He et al. 2020; Jiang et al. 2023). In addition, *Epimedium* extract (containing icariin) promotes C2C12 myotubular hypertrophy, having the potential to treat sarcopenia (Lin et al. 2021).

In a recent study, Yang et al. found a therapeutic effect of icariin on bilaterally orchiectomized (ORX)-induced sarcopenia rats (Yang et al. 2022). Oral administration of icariin increased forelimb grip strength and muscle fiber cross-sectional area in ORX-induced rats. Icariin also enhanced specific muscle strength in the flounder and toe extensor muscles and upregulated the mRNA expression of myosin heavy chain (MyHC) in these muscles (Yang et al. 2022). In a previous study, *Epimedium* extract activated C2C12 myotubular insulin-like growth factor 1 (IGF-1)/Akt/mTOR and promoted myotubular hypertrophy (Lin et al. 2021). However, in ORX-induced rats' muscles, icariin did not significantly activate Akt/mTOR signaling pathway (Yang et al. 2022). This result suggests that the effect of icariin on increasing muscle mass may not depend on the enhanced protein synthesis associated with Akt/mTOR signaling. Icariin also alleviated the conversion of type I to type II fibers in ORX skeletal muscle (Yang et al. 2022). Compared to type II fibers, type I fibers have a higher mitochondrial fusion rate, contain more mitochondria, and have higher fatigue resistance properties (Fitts 1994). Therefore, the elevating effect of icariin on type I fiber content in ORX mice may involve a mechanism to stabilize the mitochondrial state. Moreover, the expression of forkhead box class O 3a (FOXO3a), atrogin-1, and MuRF1 mRNAs, which are positively associated with protein degradation, was all downregulated after icariin intervention (Yang et al. 2022).

5,7-Dimethoxyflavone

5,7-Dimethoxyflavone (C₁₇H₁₄O₄) (DMF) is the main flavonoid component of *Kaempferia parviflora*, a family Zingiberaceae plant native to Thailand (Saokaew et al. 2017). DMF has biological activities such as apoptosis induction, antioxidant, anti-inflammatory, and anti-obesity. It can improve endometriosis, skin photoaging, obesity, and other obesity-related metabolic syndromes (Kim et al. 2012; Park et al. 2020; Song et al. 2016). *Kaempferia parviflora* extract ameliorated the fatigue-related decrease in muscle endurance and inhibited lipopolysaccharide (LPS)-induced inflammation of C2C12 in mice (Toda et al. 2016). In addition, oral supplementation with *Kaempferia parviflora* 90 to 180 mg/day significantly improved human hand grip strength (Saokaew et al. 2017). As the main pharmacologically active component of *Kaempferia parviflora*, DMF also has the potential to alleviate sarcopenia.

Kim et al. found that the regulatory effects of DMF on protein turnover, mitochondrial biogenesis, and inflammation contribute to the remission of sarcopenia in aged mice (Kim and Hwang 2020b). Force-feeding DMF enhanced grip strength and increased muscle volume, weight, and cross-sectional area in aged mice. The main signaling pathway responsible for the improved abnormal protein turnover,

PI3K/Akt/mTOR, was activated by DMF (Kim and Hwang 2020b). DMF increased the phosphorylated expression of cytoplasmic FOXO3 in aging mice skeletal muscle and inhibited the mRNA expression of MuRF1 and atrogin-1, markers of muscle atrophy. The expression of autophagy-lysosome system-related biomarkers Beclin1 and LC3 and autophagy-related proteins 4 and 7 was downregulated by DMF (Kim and Hwang 2020b). In addition, Kim et al. explored the effects of DMF on mitochondrial biogenesis pathways in flounder muscle enriched with type I muscle fibers. DMF significantly reversed the senescence-induced decrease in PGC-1 α , NRF-1, and TFAM mRNA expression (Kim and Hwang 2020b). For senescence-associated inflammatory responses, DMF reduced the levels of inflammatory cytokines IL-6 and TNF- α and inhibited the activation of NF- κ B signaling (Kim and Hwang 2020b). Notably, TNF- α produced by autocrine and paracrine pathways can further worsen aging-associated muscle atrophy by inactivating the PI3K-Akt pathway and blocking mitochondrial biogenesis (Wang et al. 2014).

Morroniside

Morroniside (C₁₇H₂₆O₁₁), a cyclic enol ether terpene, can be obtained from the traditional Chinese medicinal plant *Cornus officinalis*. Under the guidance of traditional Chinese medicine theory, *Cornus officinalis* can nourish the liver and kidneys and has been used to treat pain in the back and knees, dizziness, tinnitus, impotence, spermatorrhea, and excessive menstruation Huang et al. 2018a, b). Morroniside, as the main medicinal component of *Cornus officinalis*, has anti-inflammatory, antioxidant, and anti-apoptotic biological activities, having therapeutic potential for cardiac, skeletal, and neurological diseases (Huang et al. 2018a, b). Recently, studies have shown that morroniside is effective and safe for treating inflammatory muscle atrophy (Yi et al. 2022).

Morroniside can improve inflammatory muscle atrophy by inhibiting classical and non-classical NF- κ B signaling and regulating protein synthesis/degradation. By performing genetic and pathway genetic association analysis based on European populations, Yi et al. identified the tumor necrosis factor- α (TNF- α) gene as a candidate gene for muscle atrophy and grip strength, and TNF- α -induced activation of the NF- κ B signaling pathway was associated with muscle atrophy in humans (Yi et al. 2022). Yi et al. found that denervated muscular atrophy mice had higher mRNA and protein level expression of TNF- α in the gastrocnemius muscle than sham-operated mice (Yi et al. 2022). Morroniside intervention restored lean body mass loss in mice's tibialis anterior, toe extensor, flounder, and gastrocnemius muscles (Yi et al. 2022). In addition, denervation decreased paired box protein Pax-7 (Pax7) levels (a marker of mature muscle stem cells), the number of Pax7 + satellite cells, and the mean

cross-sectional area of muscle fibers. These changes were reversed by morroniside (Yi et al. 2022). Mechanism-wise, morroniside significantly reduced the mRNA levels of RelA, RelB, and p52, indicating that classical and non-classical NF- κ B signalings were inhibited. The levels of upregulated inflammatory factors, IL-6, IL-1b, C-reactive protein, NOD-like receptor protein 3, and prostaglandin-endoperoxide synthase 2, in denervated mice muscle, were inhibited by morroniside (Yi et al. 2022). In vitro, morroniside activated the IGF-1/IGF-1R/PI3K/Akt and bone morphogenetic protein (BMP) 14/BMPR2/activin receptor-like kinase (ALK) 2/3/Smad5/9 signaling pathways associated with protein synthesis, counteracting the inhibitory effect of TNF- α on these pathways in C2C12 myotubes (Yi et al. 2022). In vitro and in vivo, morroniside suppressed the levels of FOXO3, atrogin-1, and MuRF1 mRNAs, signaling a weakening of the muscle protein degradation pathway. Furthermore, the levels of MyHC ubiquitination, Bnip3, Lc3a, and Lc3b gene expression were downregulated by morroniside, suggesting that the ubiquitin-proteasome system and the autophagy-lysosome system may also be involved in the improvement of muscle protein degradation (Yi et al. 2022). Notably, the ubiquitin-proteasome and autophagy-lysosome pathways and muscle-specific E3 ubiquitin ligases associated with muscle protein degradation are associated with activating NF- κ B signaling and TNF- α (Li et al. 2020; Thoma and Lightfoot 2018).

Dihydromyricetin, glabridin

Ampelopsis grossedentata, dried leaves, and stems are traditional Chinese plant resources used for medicine and food (Carneiro et al. 2021). Dihydromyricetin (DHM), a flavanone from *Ampelopsis grossedentata*, has anti-inflammatory, antioxidant, anti-cancer, hepatoprotective, neuroprotective, and other biological activities (Carneiro et al. 2021). DHM has preventive and therapeutic effects on several metabolic diseases, such as non-alcoholic fatty liver, diabetes mellitus, atherosclerosis, and osteoporosis (Carneiro et al. 2021). Glabridin (C₂₀H₂₀O₄) is an isoflavone found in the root extract of *Glycyrrhiza glabra* L, a traditional plant used for medicinal and food purposes. Glabridin has many biological activities, such as antioxidant, anti-inflammatory, anti-atherosclerotic, regulating energy metabolism, skin whitening, and anti-osteoporosis (Simmler et al. 2013). It can improve cardiovascular disease, neuronal disease, and certain metabolic disorders, including obesity and diabetes (Chung et al. 2022). Both DHM and glabridin have therapeutic effects on DEXA-induced muscle atrophy.

DHM attenuated DEXA-induced muscle atrophy in rats by improving mitochondrial function and abnormal protein metabolism. In experiments performed by Huang et al., DHM infusion alleviated DEXA-induced reduction in muscle/body

weight ratio and increased grip strength in rats' anterior tibial and gastrocnemius muscles (Huang et al. 2018a, b). DHM ameliorated DEXA-induced abnormal mitochondrial morphology and increased mitochondrial swelling in the skeletal muscle of rats. DHM also restored the loss of potential in damaged mitochondrial membranes and increased mitochondrial respiratory chain complex I, II, and IV and ATP synthase activities (Huang et al. 2018a, b). According to previous information, reduced activity of mitochondrial respiratory chain complexes promoted the production of mitochondrial ROS, which in turn caused oxidative stress damage to tissues (Li et al. 2012; Yang et al. 2014). DHM reversed the elevation of DEXA-induced protein hydroxyl and MDA content and attenuated oxidative stress. In addition, the anti-DEXA-induced muscular dystrophy effect of DHM also benefited from its modulation of protein synthesis/protein degradation pathway: (1) DHM inhibited the DEXA-induced increase in FOXO3a, atrogin-1, and MuRF1 expression and reduced protein degradation; (2) DHM restored the DEXA-induced Akt/mTOR pathway inactivation and promoted protein synthesis (Huang et al. 2018a, b). Enhanced mitochondrial function has been shown to ameliorate muscle atrophy (Hyatt et al. 2019). However, whether and how the improvement of mitochondrial function by DHM affects the synthesis and degradation of muscle proteins remains to be further investigated.

Glabridin can ameliorate DEXA-induced muscle atrophy by inhibiting the binding of DEXA to glucocorticoid receptors and reducing protein degradation. After Yoshioka et al. took oral glabridin intervention in DEXA-induced mice, the trend of muscle weight loss in the tibialis anterior and gastrocnemius muscles was suppressed (Yoshioka et al. 2019). Mechanistically, glabridin prevented the elevated expression of MuRF1 and Cbl-b and the phosphorylation of FOXO3a in DEXA-induced mice muscles, which regulate protein degradation (Yoshioka et al. 2019). In addition, glabridin directly inhibited DEXA-induced nuclear translocation of the glucocorticoid receptor, possibly exerting a dual regulatory effect on protein synthesis promotion and degradation inhibition (Schakman et al. 2008; Yoshioka et al. 2019). Given the proven safety profile of *Ampelopsis grossedentata*, glabridin supplementation as a glucocorticoid receptor antagonist may become an essential alternative therapy for treating glucocorticoid-related muscular atrophy and sarcopenia.

Table 2 summarizes the therapeutic effects and mechanisms of 11 traditional herbal active ingredients mentioned above for sarcopenia (Table 2).

Conclusion and future directions

As the global population ages, sarcopenia will become a significant disease that endangers human health and causes a heavy medical and economic burden. Although

the pathogenesis is not identical, primary and secondary sarcopenia share similar pathological histological alterations. Moreover, muscle alterations in the same individual are usually influenced by multiple factors, including aging, lack of exercise, nutritional deficiencies, and chronic diseases. Therefore, an in-depth study of the pathogenesis of sarcopenia due to various factors and the search and development of safe and effective therapies based on the relevant pathogenesis is urgently needed.

Pre-clinical evidence suggests that traditional herbal active ingredients, and compounding, exert positive therapeutic effects and mechanisms for improving primary and some secondary sarcopenia. Based on the results of this review, we have drawn the following main conclusions: (1) The pathological mechanisms of the animal model of senescence-associated sarcopenia include protein synthesis/protein degradation imbalance, oxidative stress, inflammation, apoptosis, insulin resistance, impaired mitochondrial biogenesis, and exacerbation of the autophagy-lysosome system. By improving these pathological mechanisms, icariin and DMF improved muscle morphology, promoted myoblast differentiation, and enhanced grip strength in an animal model of senescence-associated sarcopenia; (2) the pathological mechanisms of the DEXA-associated sarcopenia model are similar to those of the senescence-associated model and include oxidative stress, inflammatory response, apoptosis, and protein synthesis/protein degradation imbalance. S-Rg3, DHM, and glabridin optimized some muscle morphology (muscle mass, muscle area), increased muscle strength, promoted myoblast differentiation, and inhibited muscle fibrosis in an animal model of DEXA-associated sarcopenia; (3) baicalein, CS, and DCS inhibited the activation of NF- κ B, while AI blocked the activation of STAT3. They ameliorated muscle atrophy in an animal model of tumor cachexia; (4) disruption of protein synthesis/protein degradation balance is a shared underlying critical pathological mechanism in many types of sarcopenia. PI3K/Akt/mTOR pathway activation upregulates MyoD, MHC, and myostatin, promotes protein synthesis, and improves abnormal amino acid-protein metabolism, while STAT3/PKM2/SNAP23, AMPK/FOXO3a, muscle growth inhibitor/Smad/FOXO pathway, and ubiquitin-proteasome and autophagy-lysosome systems promoted the expression of MuRF1 and atrogin-1, Cbl-b, and FOXO and accelerated the skeletal muscle in protein degradation. Overall, this review comprehensively brings together the therapeutic effects and molecular mechanisms of the world's traditional herbal active ingredients for animal models of primary and some secondary sarcopenia. The paper also covers a wealth of meaningful information on the pathophysiological mechanisms of different sarcopenia conditions. Such information will provide a comprehensive

Table 2 Therapeutic effects and mechanisms of the 11 traditional herbal active ingredients for animal models related to sarcopenia

Traditional herbal remedy	Animal and cellular models	Animal sample size	Intervention methods	Intervention dose	Duration of intervention	Therapeutic effects	Potential mechanisms	Reference
Baicalin	CT26 adenocarcinoma-induced cachexia mice	48	Intraperitoneal injection of a solution containing baicalin	50, 150 mg/kg/day	15 days	1. Delayed the loss of skeletal muscle 2. Inhibited muscle fiber degradation	3. Downregulation of TNF- α and IL-6 expression and inhibition of NF- κ B activation 4. Inhibition of MuRF1 and atrogin-1 expression	(Li et al. 2014)
Carnosol	C26 tumor-bearing mice	Not available	Intraperitoneal injection of a solution containing carnosol	10 mg/kg/day	Until the mice lose more than 10% of their body weight or when the tumor volume reaches 2000 mm ³	Improved muscle fiber atrophy	1. Regulation of NF- κ B and Akt signaling pathways 2. Inhibition of atrogin-1 expression	(Lu et al. 2021)
Atractylenolide I	C26 tumor-bearing mice	22	Intraperitoneal injection of a solution containing atractylenolide I	25 mg/kg/day	18 days	Inhibition of the reduction of muscle fiber cross-sectional area	1. Inhibited STAT3/PKM2/SNAP23 pathway 2. Decreased the levels of IL-6 and extracellular vesicles	(Fan et al. 2022a)
20(S)-ginsenoside Rg3	Older <i>Drosophila</i>	60	Consumption of foods containing 20(S)-ginsenoside Rg3	1 mg/mL/day	7 days	Inhibition of muscle atrophy	1. Activation of Akt/mTOR/FOXO3 signaling pathway 2. Promotion of the nuclear translocation of dFOXO	(Wang et al. 2020a, b)
Ginsenoside compound K	High-fat diet-induced mice	Not available	Intravenous ginsenoside compound K	1 mg/kg/day	8 weeks	Suppressed skeletal muscle cell apoptosis	1. Promoted AMPK/autophagy signaling pathway 2. Promoted p38 and Akt phosphorylation. Increased the levels of MyoD, MHC, and myogenic hormone	(Kim et al. 2022)

Table 2 (continued)

Traditional herbal remedy	Animal and cellular models	Animal sample size	Intervention methods	Intervention dose	Duration of intervention	Therapeutic effects	Potential mechanisms	Reference
Curcumin	Chronic kidney disease mice	Not available	Consuming a diet containing curcumin	100 mg/kg/day	12 weeks	1. Improvement of muscle atrophy and muscle function 2. Enhanced mitochondrial biogenesis and improved mitochondrial function	Inhibition of GSK-3 β activity 1. Decreased proteasomal pancreatic rennin-like and caspase-9 activity 2. Inhibited the increase in Smac/DIABLO protein levels 3. Increased XIAP protein levels	(Wang et al. 2020a, b)
Curcumin	Rats with unilateral hind limb immobilization	32	Curcumin injection	1 mg/kg/day	18 days	Alleviated muscle atrophy and increased cross-sectional area of muscle fibers	1. Restored muscle fiber area 2. Inhibition of fiber conversion from type I to type II 3. Reduced muscle degradation	(Vazeille et al. 2012)
Icariin	bilaterally orchiectomized rat	18	Gavage with icariin solution	50 mg/kg/day	8 weeks	1. Restored muscle fiber area 2. Inhibition of fiber conversion from type I to type II 3. Reduced muscle degradation	1. Increased expression of MyHC 2. Stabilized mitochondrial state 3. Inhibition of the expression of FOXO3a, atrogen-1, and MuRF1	(Yang et al. 2022)
5,7-Dimethoxyflavone	Aged mice	Not available	Oral 5,7-dimethoxyflavone	25, 50 mg/kg/day	8 weeks	1. Enhanced body function and grip strength 2. Increased muscle volume, weight, and cross-sectional area	1. Activated PI3K/Akt/mTOR signaling 2. Increased cytoplasmic FOXO3 phosphorylation expression. Inhibition the expression of MuRF1 and atrogen-1 3. Downregulated the expression of Beclin1, LC3, autophagy-related proteins 4 and 7 4. Upregulated PGC-1 α , NRF-1, and Tfam mRNA expression 5. Decreased TNF- α and IL-6 levels. Suppressed the activation of NF- κ B signaling	(Kim and Hwang 2020b)

Table 2 (continued)

Traditional herbal remedy	Animal and cellular models	Animal sample size	Intervention methods	Intervention dose	Duration of intervention	Therapeutic effects	Potential mechanisms	Reference
Morroniside	Denervation induced mice	24	Intraperitoneal injection of morroniside	10 or 20 µg/kg/day	12 days	1. Prevented skeletal muscle atrophy 2. Diminished inflammation in muscles	1. Inhibition of classical and non-classical NF-κB signaling 2. Promotion of IGF-1/IGF-1R/PI3K/Akt and BMP14/BMP2/ALK2/3/Smad5/9 signaling 3. Downregulation of muscle growth inhibitor, the ubiquitin-proteasome system, and autophagy-lysosome system pathways	(Yi et al. 2022)
Dihydromyricetin	Dexamethasone-induced rats	50	Feeding of distilled water containing dihydromyricetin	50, 100, 200 mg/kg/day	24 days	Improved muscle atrophy and muscle dysfunction	1. Suppressed DEXA-induced increase in FOXO3a, atrogin-1, and MuRF1 expression 2. Restored the Akt/mTOR pathway	(Huang et al. 2018a, b)
Glabridin	Dexamethasone-induced mice	12	Oral glabridin	40 mg/kg/day	2 weeks	Delayed muscle weight loss	1. Blocked MuRF1 and Cbl-b expression and inhibited FOXO3a phosphorylation 2. Inhibited dexamethasone-induced nuclear translocation of the glucocorticoid receptor	(Yoshioka et al. 2019)

and practical molecular perspective for future studies on treating primary and secondary sarcopenia with traditional herbal active ingredients.

The results of this review support that world traditional herbal active ingredients are positive and encouraging for the treatment of various sarcopenia. However, we still have some limited suggestions for the future research direction: (1) Currently, aging models, genetically engineered models, hindlimb suspension models, chemically induced or diet-induced models, degeneration models, and immobilization models are the most common and adopted animal models in the study of sarcopenia (Xie et al. 2021). However, these animal models lack accurate and accepted references for diagnostic criteria for sarcopenia. Further clarification and development of diagnostic indicators for sarcopenia in these models are needed; (2) except for aging, some chronic diseases such as oncologic cachexia, COPD, diabetes, and CKD also promote the development of sarcopenia. The number of studies on sarcopenia secondary to these common diseases is still tiny compared to those on aging-related sarcopenia. Furthermore, it is essential to note that the pathogenesis of secondary sarcopenia and primary sarcopenia partially overlaps. Therefore, exploring the pathogenesis of secondary sarcopenia is complementary and informative for the study of primary sarcopenia. In the future, relevant researchers must pay equal attention to studying the mechanisms of various secondary sarcopenia; (3) the therapeutic mechanisms of some traditional drugs and compounding for sarcopenia need further improvement and exploration. For example, DHM can alleviate DEXA-induced muscle atrophy by improving mitochondrial function. However, it is unknown whether DHM-induced relief of mitochondrial function affects signaling pathways related to protein metabolisms, such as PI3K/Akt/mTOR and AMPK /FOXO3a; (4) sarcopenia is a human pathology affected by aging, chronic diseases, nutritional status, and exercise. The pathological mechanisms are not precisely the same in different types of sarcopenia. Therefore, in the future, for different types of sarcopenia, appropriate traditional herbal active ingredients should be selected for precise intervention according to their specific pathological mechanisms.

Acknowledgements Firstly, we would like to thank Prof. Huanan Jia, from Chengdu University of Traditional Chinese Medicine Hospital, Department of Geriatrics, for reviewing the initial version of this article. Secondly, we would like to thank Kaixi Ding's parents (Xia Wang and Bing Ding), grandfather (Qingquan Ding), and uncle (Yi Ding), as they have inspired Kaixi Ding's enthusiasm and pursuit of medical research from the beginning to the present.

Authors contributions Conceptualization: Kaixi Ding and Wei Jiang. Methodology: Kaixi Ding and Wei Jiang. Software: Juejue Zhangwang and Yu Wang. Validation: Juejue Zhangwang and Yu Wang. Writing—original draft preparation: Kaixi Ding. Writing—review and editing: Kaixi Ding and Wei Jiang. Visualization: Wei Jiang. Supervision: Ming

Lei. Project administration: Ming Lei and Jing Zhang. The authors confirm that no paper mill and artificial intelligence was used.

Data availability Not applicable.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication All authors unanimously approved the publication of this manuscript.

Competing interests The authors declare no competing interests.

References

- Acharyya S, Butchbach MER, Sahenk Z et al (2005) Dystrophin glycoprotein complex dysfunction: a regulatory link between muscular dystrophy and cancer cachexia. *Cancer Cell* 8:421–432. <https://doi.org/10.1016/j.ccr.2005.10.004>
- Akaberi M, Sahebkar A, Emami SA (2021) Turmeric and curcumin: from traditional to modern medicine. *Adv Exp Med Biol* 1291:15–39. https://doi.org/10.1007/978-3-030-56153-6_2
- Attaix D, Ventadour S, Codran A et al (2005) The ubiquitin-proteasome system and skeletal muscle wasting. *Essays Biochem* 41:173–186. <https://doi.org/10.1042/EB0410173>
- Bagherniya M, Mahdavi A, Shokri-Mashhadi N et al (2022) The beneficial therapeutic effects of plant-derived natural products for the treatment of sarcopenia. *J Cachexia Sarcopenia Muscle* 13:2772–2790. <https://doi.org/10.1002/jcsm.13057>
- Barbalho SM, Flato UAP, Tofano RJ et al (2020) Physical exercise and myokines: relationships with sarcopenia and cardiovascular complications. *Int J Mol Sci* 21:3607. <https://doi.org/10.3390/ijms21103607>
- Bar-Shai M, Carmeli E, Ljubuncic P, Reznick AZ (2008) Exercise and immobilization in aging animals: the involvement of oxidative stress and NF-kappaB activation. *Free Radic Biol Med* 44:202–214. <https://doi.org/10.1016/j.freeradbiomed.2007.03.019>
- Bauer J, Morley JE, Schols AMWJ et al (2019) Sarcopenia: a time for action. An SCWD position paper. *J Cachexia Sarcopenia Muscle* 10:956–961. <https://doi.org/10.1002/jcsm.12483>
- Befroy DE, Petersen KF, Dufour S et al (2007) Impaired mitochondrial substrate oxidation in muscle of insulin-resistant offspring of type 2 diabetic patients. *Diabetes* 56:1376–1381. <https://doi.org/10.2337/db06-0783>
- Bruyère O, Beaudart C, Ethgen O et al (2019) The health economics burden of sarcopenia: a systematic review. *Maturitas* 119:61–69. <https://doi.org/10.1016/j.maturitas.2018.11.003>
- Bryner RW, Woodworth-Hobbs ME, Williamson DL, Alway SE (2012) Docosahexaenoic acid protects muscle cells from palmitate-induced atrophy. *ISRN Obes* 2012:647348. <https://doi.org/10.5402/2012/647348>
- Carneiro RCV, Ye L, Baek N et al (2021) Vine tea (*Ampelopsis grosedentata*): a review of chemical composition, functional properties, and potential food applications. *J Funct Foods* 76:104317. <https://doi.org/10.1016/j.jff.2020.104317>
- Cawthon PM, Lui L-Y, Taylor BC et al (2017) Clinical definitions of sarcopenia and risk of hospitalization in community-dwelling older men: the osteoporotic fractures in men study. *J Gerontol A Biol Sci Med Sci* 72:1383–1389. <https://doi.org/10.1093/gerona/glw327>

- Chung C-L, Chen J-H, Huang W-C et al (2022) Glabridin, a bioactive flavonoid from licorice, effectively inhibits platelet activation in humans and mice. *Int J Mol Sci* 23. <https://doi.org/10.3390/ijms231911372>
- Cruz-Jentoft AJ, Kiesswetter E, Drey M, Sieber CC (2017) Nutrition, frailty, and sarcopenia. *Aging Clin Exp Res* 29:43–48. <https://doi.org/10.1007/s40520-016-0709-0>
- Cruz-Jentoft AJ, Bahat G, Bauer J et al (2019) Sarcopenia: revised European consensus on definition and diagnosis. *Age Ageing* 48:601. <https://doi.org/10.1093/ageing/afz046>
- Deng M, Chen H, Long J et al (2021) Atractylenolides (I, II, and III): a review of their pharmacology and pharmacokinetics. *Arch Pharm Res* 44:633–654. <https://doi.org/10.1007/s12272-021-01342-6>
- Dhillon RJS, Hasni S (2017) Pathogenesis and management of sarcopenia. *Clin Geriatr Med* 33:17–26. <https://doi.org/10.1016/j.cger.2016.08.002>
- Ding K, Jiang W, Jia H, Lei M (2022) Synergistically anti-multiple myeloma effects: flavonoid, non-flavonoid polyphenols, and bortezomib. *Biomolecules* 12:1647. <https://doi.org/10.3390/biom12111647>
- Fan M, Gu X, Zhang W et al (2022a) Atractylenolide I ameliorates cancer cachexia through inhibiting biogenesis of IL-6 and tumour-derived extracellular vesicles. *J Cachexia Sarcopenia Muscle* 13:2724–2739. <https://doi.org/10.1002/jcsm.13079>
- Fan M, Sun W, Gu X et al (2022b) The critical role of STAT3 in biogenesis of tumor-derived exosomes with potency of inducing cancer cachexia in vitro and in vivo. *Oncogene* 41:1050–1062. <https://doi.org/10.1038/s41388-021-02151-3>
- Fitts RH (1994) Cellular mechanisms of muscle fatigue. *Physiol Rev* 74:49–94. <https://doi.org/10.1152/physrev.1994.74.1.49>
- Go G-Y, Lee S-J, Jo A et al (2017) Ginsenoside Rg1 from Panax ginseng enhances myoblast differentiation and myotube growth. *J Ginseng Res* 41:608–614. <https://doi.org/10.1016/j.jgr.2017.05.006>
- Günther S, Kim J, Kostin S et al (2013) Myf5-positive satellite cells contribute to Pax7-dependent long-term maintenance of adult muscle stem cells. *Cell Stem Cell* 13:590–601. <https://doi.org/10.1016/j.stem.2013.07.016>
- Guo L-T, Wang S-Q, Su J et al (2019) Baicalin ameliorates neuroinflammation-induced depressive-like behavior through inhibition of toll-like receptor 4 expression via the PI3K/AKT/FoxO1 pathway. *J Neuroinflammation* 16:95. <https://doi.org/10.1186/s12974-019-1474-8>
- He C, Wang Z, Shi J (2020) Pharmacological effects of icariin. *Adv Pharmacol* 87:179–203. <https://doi.org/10.1016/bs.apha.2019.10.004>
- Huang J, Zhang Y, Dong L et al (2018a) Ethnopharmacology, phytochemistry, and pharmacology of *Cornus officinalis* Sieb. et Zucc. *J Ethnopharmacol* 213:280–301. <https://doi.org/10.1016/j.jep.2017.11.010>
- Huang Y, Chen K, Ren Q et al (2018b) Dihydromyricetin attenuates dexamethasone-induced muscle atrophy by improving mitochondrial function via the PGC-1 α pathway. *Cell Physiol Biochem* 49:758–779. <https://doi.org/10.1159/000493040>
- Hyatt H, Deminice R, Yoshihara T, Powers SK (2019) Mitochondrial dysfunction induces muscle atrophy during prolonged inactivity: a review of the causes and effects. *Arch Biochem Biophys* 662:49–60. <https://doi.org/10.1016/j.abb.2018.11.005>
- Jiang W, Ding K, Yue R, Lei M (2023) Therapeutic effects of icariin and icariside II on diabetes mellitus and its complications. *Crit Rev Food Sci Nutr* 1–26. <https://doi.org/10.1080/10408398.2022.2159317>
- Kashyap D, Kumar G, Sharma A et al (2017) Mechanistic insight into carnosol-mediated pharmacological effects: recent trends and advancements. *Life Sci* 169:27–36. <https://doi.org/10.1016/j.lfs.2016.11.013>
- Kim C, Hwang J-K (2020a) Flavonoids: nutraceutical potential for counteracting muscle atrophy. *Food Sci Biotechnol* 29:1619–1640. <https://doi.org/10.1007/s10068-020-00816-5>
- Kim C, Hwang J-K (2020b) The 5,7-dimethoxyflavone suppresses sarcopenia by regulating protein turnover and mitochondria biogenesis-related pathways. *Nutrients* 12:1079. <https://doi.org/10.3390/nu12041079>
- Kim JK, Mun S, Kim M-S et al (2012) 5,7-Dimethoxyflavone, an activator of PPAR α/γ , inhibits UVB-induced MMP expression in human skin fibroblast cells: 5,7-DMF inhibits MMP expression. *Exp Dermatol* 21:211–216. <https://doi.org/10.1111/j.1600-0625.2011.01435.x>
- Kim TJ, Pyun DH, Kim MJ et al (2022) Ginsenoside compound K ameliorates palmitate-induced atrophy in C2C12 myotubes via promyogenic effects and AMPK/autophagy-mediated suppression of endoplasmic reticulum stress. *J Ginseng Res* 46:444–453. <https://doi.org/10.1016/j.jgr.2021.09.002>
- Lee H, Kong G, Tran Q et al (2020) Relationship between ginsenoside Rg3 and metabolic syndrome. *Front Pharmacol* 11:130. <https://doi.org/10.3389/fphar.2020.00130>
- Li X-X, He G-R, Mu X et al (2012) Protective effects of baicalin against rotenone-induced neurotoxicity in PC12 cells and isolated rat brain mitochondria. *Eur J Pharmacol* 674:227–233. <https://doi.org/10.1016/j.ejphar.2011.09.181>
- Li B, Wan L, Li Y et al (2014) Baicalin, a component of *Scutellaria baicalensis*, alleviates anorexia and inhibits skeletal muscle atrophy in experimental cancer cachexia. *Tumour Biol* 35:12415–12425. <https://doi.org/10.1007/s13277-014-2558-9>
- Li F, Li X, Peng X et al (2017) Ginsenoside Rg1 prevents starvation-induced muscle protein degradation via regulation of AKT/mTOR/FoxO signaling in C2C12 myotubes. *Exp Ther Med* 14:1241–1247. <https://doi.org/10.3892/etm.2017.4615>
- Li J, Yi X, Yao Z et al (2020) TNF receptor-associated factor 6 mediates TNF α -induced skeletal muscle atrophy in mice during aging: TRAF6 induces muscle loss during aging. *J Bone Miner Res* 35:1535–1548. <https://doi.org/10.1002/jbmr.4021>
- Lin Y-A, Li Y-R, Chang Y-C et al (2021) Activation of IGF-1 pathway and suppression of atrophy related genes are involved in Epimedium extract (icariin) promoted C2C12 myotube hypertrophy. *Sci Rep* 11:10790. <https://doi.org/10.1038/s41598-021-89039-0>
- Liu T, Zhu L, Wang L (2022) A narrative review of the pharmacology of ginsenoside compound K. *Ann Transl Med* 10:234. <https://doi.org/10.21037/atm-22-501>
- Lu S, Li Y, Shen Q et al (2021) Carnosol and its analogues attenuate muscle atrophy and fat lipolysis induced by cancer cachexia. *J Cachexia Sarcopenia Muscle* 12:779–795. <https://doi.org/10.1002/jcsm.12710>
- Ma K, Huang F, Qiao R, Miao L (2022) Pathogenesis of sarcopenia in chronic obstructive pulmonary disease. *Front Physiol* 13:850964. <https://doi.org/10.3389/fphys.2022.850964>
- Mijnarends DM, Koster A, Schols JMGA et al (2016) Physical activity and incidence of sarcopenia: the population-based AGES—Reykjavik study. *Age Ageing* 45:614–620. <https://doi.org/10.1093/ageing/afw090>
- Naseeb MA, Volpe SL (2017) Protein and exercise in the prevention of sarcopenia and aging. *Nutr Res* 40:1–20. <https://doi.org/10.1016/j.nutres.2017.01.001>
- Papadopoulou SK, Papadimitriou K, Voulgaridou G et al (2021) Exercise and nutrition impact on osteoporosis and sarcopenia—the incidence of osteosarcopenia: a narrative review. *Nutrients* 13:4499. <https://doi.org/10.3390/nu13124499>
- Park W, Park MY, Song G, Lim W (2020) 5,7-Dimethoxyflavone induces apoptotic cell death in human endometriosis cell lines by

- activating the endoplasmic reticulum stress pathway. *Phytother Res* 34:2275–2286. <https://doi.org/10.1002/ptr.6677>
- Petermann-Rocha F, Balntzi V, Gray SR et al (2022) Global prevalence of sarcopenia and severe sarcopenia: a systematic review and meta-analysis. *J Cachexia Sarcopenia Muscle* 13:86–99. <https://doi.org/10.1002/jcsm.12783>
- Pyun DH, Kim TJ, Kim MJ et al (2021) Endogenous metabolite, kynurenic acid, attenuates nonalcoholic fatty liver disease via AMPK/autophagy- and AMPK/ORP150-mediated signaling. *J Cell Physiol* 236:4902–4912. <https://doi.org/10.1002/jcp.30199>
- Reddy SS, Shruthi K, Prabhakar YK et al (2018) Implication of altered ubiquitin-proteasome system and ER stress in the muscle atrophy of diabetic rats. *Arch Biochem Biophys* 639:16–25. <https://doi.org/10.1016/j.abb.2017.12.015>
- Rhoads MG, Kandarian SC, Pacelli F et al (2010) Expression of NF-kappaB and IkappaB proteins in skeletal muscle of gastric cancer patients. *Eur J Cancer* 46:191–197. <https://doi.org/10.1016/j.ejca.2009.10.008>
- Rondanelli M, Miccono A, Peroni G et al (2016) A systematic review on the effects of botanicals on skeletal muscle health in order to prevent sarcopenia. *Evid Based Complement Alternat Med* 2016:5970367. <https://doi.org/10.1155/2016/5970367>
- Roshanravan B, Gamboa J, Wilund K (2017) Exercise and CKD: skeletal muscle dysfunction and practical application of exercise to prevent and treat physical impairments in CKD. *Am J Kidney Dis* 69:837–852. <https://doi.org/10.1053/j.ajkd.2017.01.051>
- Sabatino A, Cuppari L, Stenvinkel P et al (2021) Sarcopenia in chronic kidney disease: what have we learned so far? *J Nephrol* 34:1347–1372. <https://doi.org/10.1007/s40620-020-00840-y>
- Saokaew S, Wilairat P, Raktanyakan P et al (2017) Clinical effects of Krachaidum (Kaempferia parviflora): a systematic review. *J Evid Based Complement Altern Med* 22:413–428. <https://doi.org/10.1177/2156587216669628>
- Schakman O, Gilson H, Thissen JP (2008) Mechanisms of glucocorticoid-induced myopathy. *J Endocrinol* 197:1–10. <https://doi.org/10.1677/JOE-07-0606>
- Seok YM, Yoo J-M, Nam Y et al (2021) Mountain ginseng inhibits skeletal muscle atrophy by decreasing muscle RING finger protein-1 and atrogin1 through forkhead box O3 in L6 myotubes. *J Ethnopharmacol* 270:113557. <https://doi.org/10.1016/j.jep.2020.113557>
- Shen J, Cheng J, Zhu S et al (2019) Regulating effect of baicalin on IKK/IkB/NF-kB signaling pathway and apoptosis-related proteins in rats with ulcerative colitis. *Int Immunopharmacol* 73:193–200. <https://doi.org/10.1016/j.intimp.2019.04.052>
- Sieber CC (2019) Malnutrition and sarcopenia. *Aging Clin Exp Res* 31:793–798. <https://doi.org/10.1007/s40520-019-01170-1>
- Simmler C, Pauli GF, Chen S-N (2013) Phytochemistry and biological properties of glabridin. *Fitoterapia* 90:160–184. <https://doi.org/10.1016/j.fitote.2013.07.003>
- Singh S, Meena A, Luqman S (2021) Baicalin mediated regulation of key signaling pathways in cancer. *Pharmacol Res* 164:105387. <https://doi.org/10.1016/j.phrs.2020.105387>
- Song Y, Kim M-B, Kim C et al (2016) 5,7-Dimethoxyflavone attenuates obesity by inhibiting adipogenesis in 3T3-L1 adipocytes and high-fat diet-induced obese C57BL/6J mice. *J Med Food* 19:1111–1119. <https://doi.org/10.1089/jmf.2016.3800>
- Steffl M, Sima J, Shiells K, Holmerova I (2017) The increase in health care costs associated with muscle weakness in older people without long-term illnesses in the Czech Republic: results from the Survey of Health, Ageing and Retirement in Europe (SHARE). *Clin Interv Aging* 12:2003–2007. <https://doi.org/10.2147/cia.s150826>
- Takemoto Y, Inaba S, Zhang L et al (2017) An herbal medicine, Goshajinki-gan (GJG), increases muscle weight in severe muscle dystrophy model mice. *Clin Nutr Exp* 16:13–23. <https://doi.org/10.1016/j.yclnex.2017.08.003>
- Theeuwes WF, Gosker HR, Langen RCJ et al (2017) Inactivation of glycogen synthase kinase-3 β (GSK-3 β) enhances skeletal muscle oxidative metabolism. *Biochim Biophys Acta Mol Basis Dis* 1863:3075–3086. <https://doi.org/10.1016/j.bbadis.2017.09.018>
- Thoma A, Lightfoot AP (2018) NF-kB and inflammatory cytokine signalling: role in skeletal muscle atrophy. *Adv Exp Med Biol* 1088:267–279. https://doi.org/10.1007/978-981-13-1435-3_12
- Toda K, Hitoe S, Takeda S, Shimoda H (2016) Black ginger extract increases physical fitness performance and muscular endurance by improving inflammation and energy metabolism. *Heliyon* 2:e00115. <https://doi.org/10.1016/j.heliyon.2016.e00115>
- Traub J, Bergheim I, Eibisberger M, Stadlbauer V (2020) Sarcopenia and liver cirrhosis-comparison of the European Working Group on Sarcopenia criteria 2010 and 2019. *Nutrients* 12:547. <https://doi.org/10.3390/nu12020547>
- Vazeille E, Slimani L, Claustre A et al (2012) Curcumin treatment prevents increased proteasome and apoptosome activities in rat skeletal muscle during reloading and improves subsequent recovery. *J Nutr Biochem* 23:245–251. <https://doi.org/10.1016/j.jnutbio.2010.11.021>
- Wang D-T, Yin Y, Yang Y-J et al (2014) Resveratrol prevents TNF- α -induced muscle atrophy via regulation of Akt/mTOR/FoxO1 signaling in C2C12 myotubes. *Int Immunopharmacol* 19:206–213. <https://doi.org/10.1016/j.intimp.2014.02.002>
- Wang D, Sun H, Song G et al (2018a) Resveratrol improves muscle atrophy by modulating mitochondrial quality control in STZ-induced diabetic mice. *Mol Nutr Food Res* 62:1700941. <https://doi.org/10.1002/mnfr.201700941>
- Wang D, Yang Y, Zou X et al (2020) Curcumin ameliorates CKD-induced mitochondrial dysfunction and oxidative stress through inhibiting GSK-3 β activity. *J Nutr Biochem* 83:108404. <https://doi.org/10.1016/j.jnutbio.2020.108404>
- Wang M, Ren J, Chen X et al (2020) 20(S)-Ginsenoside Rg3 promotes myoblast differentiation and protects against myotube atrophy via regulation of the Akt/mTOR/FoxO3 pathway. *Biochem Pharmacol* 180:114145. <https://doi.org/10.1016/j.bcp.2020.114145>
- Wang J, Zeng L, Zhang Y et al (2022) Pharmacological properties, molecular mechanisms and therapeutic potential of ginsenoside Rg3 as an antioxidant and anti-inflammatory agent. *Front Pharmacol* 13:975784. <https://doi.org/10.3389/fphar.2022.975784>
- Wang D, Wei L, Yang Y, Liu H (2018b) Dietary supplementation with ketoacids protects against CKD-induced oxidative damage and mitochondrial dysfunction in skeletal muscle of 5/6 nephrectomised rats. *Skelet Muscle* 8. <https://doi.org/10.1186/s13395-018-0164-z>
- Wiedmer P, Jung T, Castro JP et al (2021) Sarcopenia - molecular mechanisms and open questions. *Ageing Res Rev* 65:101200. <https://doi.org/10.1016/j.arr.2020.101200>
- Woodworth-Hobbs ME, Perry BD, Rahnert JA et al (2017) Docosahexaenoic acid counteracts palmitate-induced endoplasmic reticulum stress in C2C12 myotubes: impact on muscle atrophy. *Physiol Rep* 5:e13530. <https://doi.org/10.14814/phy2.13530>
- Xie W-Q, He M, Yu D-J et al (2021) Mouse models of sarcopenia: classification and evaluation. *J Cachexia Sarcopenia Muscle* 12:538–554. <https://doi.org/10.1002/jcsm.12709>
- Xu M, Li X, Song L (2020) Baicalin regulates macrophages polarization and alleviates myocardial ischemia/reperfusion injury via inhibiting JAK/STAT pathway. *Pharm Biol* 58:655–663. <https://doi.org/10.1080/13880209.2020.1779318>

- Yang K-C, Bonini MG, Dudley SC Jr (2014) Mitochondria and arrhythmias. *Free Radic Biol Med* 71:351–361. <https://doi.org/10.1016/j.freeradbiomed.2014.03.033>
- Yang W-Y, Cao H-J, Li L et al (2022) A phytomolecule icariin protects from sarcopenia partially by suppressing myosin heavy chain degradation in orchietomized rats. *Adv Biol (Weinh)* 6:e2200162. <https://doi.org/10.1002/adbi.202200162>
- Yi X, Tao J, Qian Y et al (2022) Morroniside ameliorates inflammatory skeletal muscle atrophy via inhibiting canonical and non-canonical NF- κ B and regulating protein synthesis/degradation. *Front Pharmacol* 13:1056460. <https://doi.org/10.3389/fphar.2022.1056460>
- Yoshioka Y, Kubota Y, Samukawa Y et al (2019) Glabridin inhibits dexamethasone-induced muscle atrophy. *Arch Biochem Biophys* 664:157–166. <https://doi.org/10.1016/j.abb.2019.02.006>
- Yuan H, Ma Q, Ye L, Piao G (2016) The traditional medicine and modern medicine from natural products. *Molecules* 21:559. <https://doi.org/10.3390/molecules21050559>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.